time: we can say that two events are simultaneous only relatively to a given motion (see here). Quantum mechanics extends this relativity in a radical way: *all* variable aspects of an object exist only in relation to other objects. It is only in interactions that nature draws the world.

In the world described by quantum mechanics there is no reality except in the *relations* between physical systems. It isn't things that enter into relations but, rather, relations that ground the notion of 'thing'. The world of quantum mechanics is not a world of objects: it is a world of events. Things are built by the happening of elementary events: as the philosopher Nelson Goodman wrote in the 1950s, in a beautiful phrase, 'An object is a monotonous process.' A stone is a vibration of quanta that maintains its structure for a while, just as a marine wave maintains its identity for a while before melting again into the sea.

What is a wave, which moves on water without carrying with it any drop of water? A wave is not an object, in the sense that it is not made of matter that travels with it. The atoms of our body, as well, flow in and away from us. We, like waves and like all objects, are a flux of events; we are processes, for a brief time monotonous …

Quantum mechanics does not describe objects: it describes processes and events which are junction points between processes.

To summarize, quantum mechanics is the discovery of three features of the world:

- *Granularity* (figure 4.8). The information in the state of a system is finite, and limited by Plank's constant.
- *Indeterminacy*. The future is not determined unequivocally by the past. Even the more rigid regularities we see are, ultimately, statistical.
- *Relationality*. The events of nature are always interactions. All events of a system occur in relation to another system.

Quantum mechanics teaches us not to think about the world in terms of 'things' which are in this or that state but in terms of 'processes' instead. A process is the passage from one interaction to another. The properties of 'things' manifest themselves in a *granular* manner only in the moment of interaction, that is to say, at the edges of the processes, and are such only in *relation* to other things. They cannot be predicted in an unequivocal way but only in a *probabilistic* one.